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Method for preparing galacto-oligosaccharides from lactulose

Abstract

Process for the preparation of a galacto-oligosaccharide preparation, which process comprises the step of contacting a lactulose-containing feed with a beta-galactosidase derived from *Papiliotrema terrestris*. The resulting galacto-oligosaccharide is acceptable to subjects suffering from GOS-related allergy and subjects having lactose intolerance.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national phase application of International Application No. PCT/EP2020/084190, filed Dec. 2, 2020, which claims benefit from European Application 19213166.2, filed Dec. 3, 2019, which are each hereby incorporated herein by reference in their entirety.

FIELD

(2) The present invention relates to the preparation of galacto-oligosaccharides (GOS) acceptable to subjects suffering from GOS-related allergy and subjects having lactose intolerance.

BACKGROUND

(3) Conventional GOS comprises a chain of galactose units and a terminal glucose unit, that arises through consecutive transgalactosylation reactions, catalyzed by a beta-galactosidase. Some of the GOS components exist naturally in human breast milk and bovine colostrum. Typical GOS preparations mainly comprise di- to hexa-saccharides.

(4) Various physiological functions of GOS have been reported, including the capacity to stimulate the growth of bifidogenic bacteria in the gut, to support normal gut transit, to contribute to natural defenses, to enhance mineral absorption, and to stimulate immune functions and lower inflammations. GOS has received particular attention for its prebiotic effects that promote the growth of *Bifidobacterium*, *Lactobacillus*, and other enteric bacteria. Therefore, GOS is commonly used in infant formula, beverages fermented by *Lactobacillus*, yogurts, juices and drinks. Some of these GOS-containing foods are certified as Food for Specified Health Uses by the Consumer Affairs Agency in Japan, and GOS is certified as Generally Recognized As Safe (GRAS) substances by the U.S. Food and Drug Administration (GRAS Notices: GRN 233, 236, 285, 286, 334, 484, 489, 495, 518, and 569).

(5) GOS is conventionally made by contacting a lactose-containing feed with a beta-galactosidase. The resulting GOS is a mixture of galacto-oligosaccharides with different degrees of polymerization (DP), including lactose. A large part of the world-wide population above 3 years of age suffers from lactose intolerance, which may result in abdominal pain, bloating, diarrhea, gas, and nausea upon consumption of lactose-containing compositions. Conventional GOS contains lactose and thus may cause these symptoms.

(6) Therefore, a need exists to produce a GOS that does not contain lactose. Solving this problem by removing lactose from conventional GOS is not a viable option because the removal of lactose requires the use of a lactose-hydrolyzing enzyme—lactase—that will not only hydrolyse lactose but also other GOS components, resulting in the loss of GOS functionalities.

(7) The beta-galactosidase enzymes that are used for the production of conventional GOS are those produced in many microorganisms such as *Bacillus circulans*, *Aspergillus oryzae*, *Kluyveromyces marxianus*, *Kluyveromyces fragilis*, *Sporobolomyces singularis*, and *Lactobacillus fermentum*. Beta-galactosidases differ in their three-dimensional structures, resulting in stereo- and regioselective formation of the glycosidic bonds. For example, a fungal beta-galactosidase derived from *Aspergillus* predominantly produces (31-6 bonds (thus resulting in a GOS preparation that predominantly comprises (31-6 bonds, which may be referred to as “6’-GOS”), while a bacterial beta-galactosidase derived from *Bacillus circulans* predominantly produce (31-4 bonds (resulting in a GOS preparation that predominantly comprises (31-4 bonds, which may also be referred to as “4’-GOS”). Moreover, beta-galactosidase produced by *B. circulans* possesses particularly strong transgalactosylation activity. As a result, GOS prepared by *B. circulans* is sold worldwide.

(8) Since its introduction to the market (1999), approximately more than 100 million of infants have consumed infant formula containing GOS prepared by *B. circulans*. It has been proven to be a safe ingredient, with a GRAS status acknowledged by the FDA. In the past few years, however, a small number of very rare cases of GOS-related allergy (~2 per million) has been reported in South East Asia. Research has shown that certain oligosaccharide structures present in GOS can exert an

allergic response in very sensitive subjects.

DETAILED DESCRIPTION

(9) The object of the present invention is therefore to provide a GOS preparation that is well tolerated by subjects suffering from lactose intolerance and by subjects suffering from allergic responses to conventional GOS obtained by *Bacillus circulans* beta-galactosidase or *Aspergillus oryzae* beta-galactosidase.

(10) This object has been met by the present invention, which involves the production of a GOS preparation by contacting a lactulose-containing feed with β -galactosidase derived from the microorganism *Papiliotrema terrestris*.

(11) A previous name of the microorganism "*Papiliotrema terrestris*" is "*Cryptococcus Papiliotrema terrestris*". The names "*Cryptococcus terrestris*" (*C. terrestris*) and "*Papiliotrema terrestris*" (*P. terrestris*) thus refer to the same organism.

(12) By " β -galactosidase derived from *P. terrestris*" is meant a β -galactosidase enzyme produced by a microorganism (of either a wild-type strain or a mutant strain) which is classified into *Papiliotrema terrestris*, or a β -galactosidase enzyme obtained by genetic engineering procedures using the β -galactosidase gene from a microorganism (of either a wild-type strain or a mutant strain) which is classified into *Papiliotrema terrestris*. Therefore, the term " β -galactosidase derived from *Papiliotrema terrestris*" also encompasses a recombinant enzyme that is produced by a host microorganism into which the β -galactosidase gene (or a modified gene thereof) obtained from *Papiliotrema terrestris* has been introduced.

(13) The resulting GOS preparation differs from conventional GOS, which is made from lactose, in that it has a fructose instead of a glucose residue at the reducing end. We therefore also refer to this lactulose-derived GOS as fGOS.

(14) This fGOS is clinically lactose-free, meaning that the ratio oligosaccharides other than lactose to lactose is at least 10. Preferably, the fGOS obtained by the process of the present invention is essentially free of lactose, meaning that it does not contain more than traces—i.e. not more than 1 wt %, preferably not more than 0.5 wt %, and most preferably not more than 0.1 wt %, based on dry matter—of lactose. In a most preferred embodiment, the fGOS obtained according to the present invention does not contain any lactose.

(15) Furthermore, the process according to the present invention allows to obtain an fGOS preparation that—without any purification or concentration steps to remove lactulose or monosugars—contains, based on dry matter, 30-60 wt %, preferably 40-60 wt % of galacto-oligosaccharide. This content of oligosaccharide does not include lactulose, but does include DP2 oligosaccharides other than lactulose. The other ingredients of this fGOS preparation mainly consist of lactulose and monosugars (fructose, galactose, glucose).

(16) The production of GOS from lactulose is known from the prior art. For instance, C. Guerrero et al., *Food Chemistry* 138 (2013) 2225-2232, discloses the production of GOS from lactose and lactulose using β -galactosidases derived from three different sources. It was found that the GOS yield from lactulose was highest with *Aspergillus oryzae*-derived beta-galactosidase and lowest with *Bacillus circulans*-derived beta-galactosidase. The reverse order was found when starting from lactose: highest yield was obtained with *Bacillus circulans*-derived beta-galactosidase; lowest yield with *Aspergillus oryzae*-derived beta-galactosidase.

(17) As shown in the Examples below, it has now been found that *P. terrestris*-derived beta-galactosidase is even better suited for GOS production from lactulose than *Aspergillus oryzae*-derived beta-galactosidase.

(18) Furthermore, the fGOS preparation obtained with the process of the present invention has a reduced (IgE-mediated) allergic response in a subject. In other words: this fGOS preparation, when administered to a subject suffering from at least one type of GOS-related allergy, i.e. an allergy caused by GOS produced by *Bacillus circulans*-derived beta-galactosidase and/or by GOS produced by *Aspergillus oryzae*-derived beta-galactosidase, evokes a reduced allergic response

when compared to a GOS preparation produced by *Bacillus circulans* or *Aspergillus oryzae*-derived beta-galactosidase. More in particular, the fGOS preparation according to the present invention has a decreased score in a Skin Prick Test in the subject and/or in a Basophil Activation Test performed on a blood sample isolated from the subject when compared to a GOS preparation obtained by *Bacillus circulans* or *Aspergillus oryzae* derived beta-galactosidase.

(19) And since the fGOS preparation is clinically lactose free, it is also tolerated well by subjects suffering from lactose intolerance.

(20) The invention therefore also relates to a method for at least partially preventing hypersensitivity to a GOS preparation in a subject, comprising administering to said subject the fGOS preparation according to the present invention or a nutritional composition comprising said fGOS preparation.

(21) The subject is a mammal, in particular a human being. Although the subject may have any age, the subject is preferably aged at least 18 months, preferably at least 24 months, even more preferably at least 3 years (36 months), and most preferably at least 13 years.

(22) The rare GOS-related allergy has not been reported in subjects having an age of 18 months or below and lactose intolerance generally does not occur below the age of 2 years.

(23) In view of the localized incidence of the 4'-GOS and/or 6'-GOS-related allergies in South East Asia (e.g. Singapore, Japan) and the abundant lactose intolerance within the Asian population, the subject is preferably of (South East) Asian origin.

(24) The β -galactosidase enzyme used for the manufacture of the fGOS preparation is known per se from patent application US 2019-119662 (originating from. PCT/JP2016/089001).

(25) In WO2019/002304, the use of this enzyme for the production of hypoallergenic GOS is disclosed. Said hypoallergenic GOS is obtained from a conventional lactose feed and may therefore cause health issues in lactose intolerant subjects.

(26) The fGOS preparation can be administered to a subject in the form of a nutritional composition. Such nutritional composition comprises (i) the fGOS preparation obtainable by the process of the present invention and (ii) at least one further ingredient selected from the group consisting of protein sources, probiotics, lipid sources, and carbohydrates.

(27) As used herein, a nutritional composition refers to any composition or formulation that goes into the alimentary canal for nutritional purposes, in whatever solid, liquid, or gaseous state. Thus, a nutritional composition can be a food item or a drink item. Examples of nutritional compositions according to the present invention are infant formula, Growing Up Milk (GUM), beverages fermented by *Lactobacillus*, yogurts, food supplements, and nutritional fortifiers.

(28) Examples of protein sources that may be present in the nutritional composition include whey proteins (e.g. whey protein concentrate or whey protein isolate), casein (e.g. micellar casein isolate), milk protein concentrate or isolate, and/or plant proteins such as soy protein. In a preferred embodiment, the protein source is a hypoallergenic or non-allergenic protein source. This includes protein hydrolysates that can be administered to subjects having intolerance against dietary proteins, more particularly cow's milk proteins, without inducing allergic reactions. Examples of such protein hydrolysates are hydrolyzed whey proteins containing hydrolysis residues having a molecular weight below 10,000 Da and casein hydrolysate with peptides of maximally 3000 Da.

(29) Examples of carbohydrate sources that may be present in the nutritional composition are disaccharides such as saccharose, monosaccharides, such as glucose, and maltodextrins, starch and carbohydrate sources having a prebiotic effect. The presence of lactose is evidently undesired.

(30) Examples of lipid sources that may be present in the nutritional composition are tri-, di-, and monoglycerides, phospholipids, sphingolipids, fatty acids, and esters or salts thereof. The lipids may have an animal, vegetable, microbial or synthetic origin. Of particular interest are polyunsaturated fatty acids (PUFAs) such as gamma linolenic acid (GLA), dihomogamma linolenic acid (DHGLA), arachidonic acid (AA), stearidonic acid (SA), eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA), docosapentaenoic acid (DPA) and conjugated linoleic acid

(CLA). CLA is important in the protection against eczema and respiratory diseases in children. This particularly involves the cis-9, trans-11 and cis-12 isomers of CLA. Examples of suitable vegetable lipid sources include sun flower oil, high oleic sun flower oil, coconut oil, palm oil, palm kernel oil, soy bean oil, etc. Examples of suitable lipid sources of animal origin include milkfat, for example anhydrous milkfat (AMF), cream, etc. In a preferred embodiment, a combination of milkfat and lipids of vegetable origin are used.

(31) The nutritional composition may further comprise a probiotic. In the context of the present invention, the term “probiotic” refers to a strain of probiotic bacteria. Probiotic bacteria are known in the art. Suitably, the probiotic bacteria are not genetically modified. Suitable probiotic bacteria include bacteria of the genus *Bifidobacteria* (e.g. *B. breve*, *B. longum*, *B. infantis*, *B. bifidum*), *Lactobacillus* (e.g. *L. Acidophilus*, *L. paracasei*, *L. johnsonii*, *L. plantarum*, *L. reuteri*, *L. rhamnosus*, *L. casei*, *L. lactis*), and *Streptococcus* (e.g. *S. thermophilus*). *B. breve* and *B. longum* are especially suitable probiotics. Suitable *B. breve* strains may for example be isolated from the faeces of healthy human milk-fed infants. The combination of a prebiotic and a probiotic is also referred to as a “synbiotic”. The probiotic may be present in the composition at any suitable concentration, suitably in a therapeutically effective amount or “amount effective for treating” in the context of the invention. Suitably, the probiotic is included in the present composition in an amount of 10²-10¹³ cfu per g dry weight of the composition, suitably 10⁵-10¹² cfu/g, most suitably 10⁷-10¹⁰ cfu/g.

(32) Further, the nutritional composition may contain one or more conventional micro ingredients, such as vitamins, antioxidants, minerals, free amino acids, nucleotides, taurine, carnitine and polyamines. Examples of suitable antioxidants are BHT, ascorbyl palmitate, vitamin E, alpha and beta carotene, lutein, zeaxanthin, lycopene and phospholipids.

(33) The β -galactosidase used in the process of the present invention has been disclosed extensively in WO2019/002304. It may be obtained from the *Papiliotrema terrestris* strain MM13-F2171 or from its mutant strains M2 and M6. Mutant strains (M2 and M6) can be obtained from *Papiliotrema terrestris* strain MM13-F2171 by means of mutagenesis with UV treatment. *Papiliotrema terrestris* strains MM13-F2171 and M2 have been deposited at a depository, as described below, and are readily available.

(34) <*Papiliotrema terrestris* Strain MM13-F2171>

(35) Depository: Patent Microorganisms Depository, National Institute of Technology and Evaluation (Room 122, 2-5-8 Kazusa Kamatari, Kisarazu-shi, Chiba, 292-0818, JAPAN).

Identification reference: *Cryptococcus terrestris* MM13-F2171. Date of deposit: Dec. 10, 2015.

Accession number: NITE BP-02177;

<*Papiliotrema terrestris* Strain M2> Depository: Patent Microorganisms Depository, National Institute of Technology and Evaluation (Room 122, 2-5-8 Kazusa Kamatari, Kisarazu-shi, Chiba, 292-0818, JAPAN). Identification reference: *Cryptococcus terrestris* APC-6431. Date of deposit: Dec. 10, 2015. Accession number: NITE BP-02178

(36) Accordingly, in one embodiment the enzyme used in the present invention is derived from *Papiliotrema terrestris* strain MM13-F2171 (Accession Number: NITE BP-02177) or APC-6431 (Accession Number: NITE BP-02178).

(37) In a further embodiment, the enzyme comprises an amino acid sequence according to any of SEQ ID NO: 1, 2, 3 or 4, or an amino acid sequence that is at least 80%, preferably at least 85%, more preferably at least 90%, even more preferably at least 95%, yet even more preferably at least 97%, and most preferably at least 99% identical to any of SEQ ID NO: 1, 2, 3 or 4.

(38) US 2019-119662 describes three kinds of β -galactosidase produced by mutant strains derived from the *Papiliotrema* microorganism (mutant strain enzymes 1, 2, and 3), and determined their amino acid sequences. These three β -galactosidase enzymes were found to have a partial sequence of the full-length amino acid sequence of the wild-type strain enzyme (the wild-type strain enzyme is shown in SEQ ID NO: 1), which is deduced from its gene sequence. Specifically, these mutant

enzymes are one having an amino acid sequence in which the N-terminal 130 amino acid residues of the full-length amino acid sequence of the wild-type strain enzyme are deleted, which is referred to as “mutant strain enzyme 1” for the purpose of description (see SEQ ID NO: 2); one having an amino acid sequence in which the N-terminal 136 amino acid residues of the full-length amino acid sequence of the wild-type strain enzyme are deleted (see SEQ ID NO:3), which is referred to as “mutant strain enzyme 2” for the purpose of description; and one having an amino acid sequence in which the N-terminal 141 amino acid residues of the full-length amino acid sequence of the wild-type strain enzyme are deleted (see SEQ ID NO:4), which is referred to as “mutant strain enzyme 3”.

(39) The term “equivalent amino acid sequence” in this case means an amino acid sequence which is partially different from the reference amino acid sequence (i.e. amino acid sequence of any one of SEQ ID NOs:1 to 4), but the difference does not substantially influence the function of the protein (beta-galactosidase activity). Thus, an enzyme having a polypeptide chain of the equivalent amino acid sequence shows a beta-galactosidase activity.

(40) The term “partial difference in the amino acid sequence” typically means mutation (change) in the amino acid sequence caused by deletion or substitution of one to several (up to, for example, 3, 5, 7, or 10) amino acids composing the amino acid sequence, or addition, insertion, or a combination thereof of one to several (up to, for example, 3, 5, 7, or 10) amino acids. The difference in the amino acid sequence is acceptable as long as the beta-galactosidase activity is maintained (the activity may be varied to a degree). As long as the conditions are satisfied, the position of the difference in the amino acid sequence is not particularly limited, and the difference may arise in a plurality of positions. The term “plurality” means, for example, a number corresponding to less than about 20%, preferably less than about 15%, more preferably less than about 10%, even more preferably less than about 5% of the total amino acids, and most preferably less than about 1%. More specifically, the equivalent protein has, for example, about 80% or more, preferably about 85% or more, more preferably about 90% or more, much more preferably about 95% or more, even more preferably about 97% or more, and most preferably about 99% or more identity with the reference amino acid sequence.

(41) The difference of the amino acid sequence may arise in a plurality of positions. Preferably, the equivalence protein is obtained by causing conservative amino acid substitution in an amino acid residue which is not essential for beta-galactosidase activity. The term “conservative amino acid substitution” means the substitution of an amino acid residue with another amino acid residue having a side chain with similar properties.

(42) Amino acid residues are classified into several families according to their side chains, such as basic side chains (for example, lysine, arginine, and histidine), acidic side chains (for example, aspartic acid and glutamic acid), uncharged polar side chains (for example, glycine, asparagine, glutamine, serine, threonine, tyrosine, and cysteine), nonpolar side chains (for example, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, and tryptophan), β -branched side chains (for example, threonine, valine, and isoleucine), and aromatic side chains (for example, tyrosine, phenylalanine, tryptophan, and histidine). Conservative amino acid substitution is preferably the substitution between amino acid residues in one family.

(43) The identity (%) between two amino acid sequences or two nucleic acid sequences (hereinafter, the term “two sequences” are used for representing either of two sequences) can be determined by the following procedure. Firstly, two sequences are aligned for optimum comparison of the two sequences (for example, a gap may be introduced into the first sequence so as to optimize the alignment with respect to the second sequence). When a molecule (amino acid residue or nucleotide) at a specific position in the first sequence and a molecule in the corresponding position in the second sequence are the same as each other, the molecules in the positions are defined as being identical. The identity between two sequences is a function of the number of identical positions shared by the two sequences (i.e., $\text{identity (\%)} = 100 \times \text{number of identical}$

positions/total number of positions). Preferably, the number and size of the gaps, which are required to optimize the alignment of the two sequences, are taken into consideration. For further information concerning the determination of the identity between two sequences, it is referred to WO 2019/002304, page 17, line 11 through page 18, line 7.

(44) An enzyme for use in the process of the present invention having the above-described amino acid sequence may also be prepared by a genetic engineering technique. For example, an appropriate host cell (for example, *Escherichia coli*) is transformed by a DNA encoding the present enzyme, and the protein expressed in the transformant is collected, and thereby preparing the present enzyme. The collected protein is treated as appropriate according to the intended use. The enzyme thus obtained as a recombinant protein may be subjected to various modifications. For example, the enzyme composed of a recombinant protein linked to any peptide or protein can be obtained by producing a recombinant protein using a vector into which a DNA encoding the enzyme has been inserted together with other appropriate DNA. In addition, modification for causing addition of a sugar chain and/or a lipid, or N- or C-terminal processing may be carried out. These modifications allow, for example, extraction of a recombinant protein, simplification of purification, or addition of biological functions. As described in US 2019-119662, an enzyme for use in the process of the present invention is advantageously produced by a transformant into which the recombinant DNA encoding the beta-galactosidase (EC 3.2.1.23) derived from *P. terrestris* is introduced, such that the gene exists as an exogenous molecule. Preferably, the transformant is prepared by transfection or transformation using the vector mentioned above. The host cell is not particularly limited as long as the present enzyme can be expressed, and it can be selected from, for example, *Bacillus* genus bacteria (e.g. *Bacillus subtilis*, *Bacillus licheniformis*, *Bacillus circulans*, etc.), lactic acid bacteria (e.g. *Lactococcus*, *Lactobacillus*, *Streptococcus*, *Leuconostoc*, *Bifidobacterium*, etc.), other bacteria (e.g. *Escherichia*, *Streptomyces*, etc.), yeast (e.g. *Saccharomyces*, *Kluyveromyces*, *Candida*, *Torula*, *Torulopsis*, *Pichia*, *Schizosaccharomyces*, etc.), and filamentous fungi (Eumycetes) (e.g. *Aspergillus* genus fungi such as *Aspergillus oryzae* and *Aspergillus niger*, *Penicillium* genus fungi, *Trichoderma* genus fungi, *Fusarium* genus fungi, etc.).

(45) The process according to the present invention involves contacting a lactulose-containing feed with the beta-galactosidase at a preferred temperature of about 50-75° C., more preferably about 63-73° C., more preferably about 65-70° C.

(46) The lactulose-containing feed preferably is an aqueous lactulose syrup comprising 40-58 wt. % lactulose, more preferably 45-55 wt. % lactulose, most preferably 50-55 wt % lactulose.

(47) The pH of the lactulose-containing feed is preferably in the range 3.5-6.5, more preferably 4.5-6.0, and most preferably 5-5.5. The pH can be regulated by a food grade buffer, such as a citrate buffer, preferably in a concentration of 0.5 mM-10 mM.

(48) The enzyme is used in a preferred concentration of at 1-20 LU/gram lactulose, more preferably 1-10 LU/gram lactulose, even more preferably of 2-8 LU/gram lactulose, and most preferably 3-6 LU/gram lactulose. Lower or higher concentrations can also be used, depending on the reaction temperature and reaction time: higher reaction temperature and/or longer reaction time allow lower enzyme concentrations.

(49) The enzyme can be used in powder form (e.g. freeze dried, vacuum dried, or spray dried) or liquid form (e.g. dissolved in a phosphoric acid buffer solution, a triethanol amine buffer solution, a tris-hydrochloric acid buffer solution, or a GOOD buffer solution).

(50) In a specific embodiment, the enzyme is used in immobilized form. Various ways of enzyme immobilization are known in the art. They typically comprise a porous carrier onto which the beta-galactosidase is immobilized via covalent binding, via physical absorption (charge-charge or *van der Waals* interaction), via gel encapsulation, or a combination thereof. Besides, carrier-free immobilized enzymes such as CLEC (cross-linked enzyme crystals) or CLEA (crosslinked enzyme aggregates) might be also applied.

- (51) Carriers that can promote direct covalent binding of the enzyme are preferred, in view of their ease of operation and absence of leakage into the reaction mixture. An example of a solid carrier is an activated acrylic polymer, preferably a functionalized polymethacrylate matrix. For example, a hexamethylenamino-functionalized polymethacrylate matrix (Sepabeads) or a microporous acrylic epoxy-activated resin, like Eupergit C 250 L, can be used.
- (52) The use of immobilized enzyme allows a repeated batch operating system involving several consecutive batches ('cycles') of GOS synthesis. It also allows for recycling of enzyme, which enables semi-continuous operation and multiple reuse of the enzyme.
- (53) The fGOS that is obtained from the process of the present invention can be isolated and purified using conventional methods, using, e.g., nanofiltration or sequential simulated moving bed (SSMB).

Description

EXAMPLES

(1) Beta-galactosidases of the following sources were used in the experiments: *Aspergillus Oryzae* (Lactase 14 DS, ex-Amano Enzyme), *Aspergillus Oryzae* (Tolerase 100, ex-DSM) *Aspergillus Oryzae* (Biolactase F, ex-Kerry Bioscience) *Papiliotrema terrestris* (β -galactosidase PT, ex-Amano Enzyme)

(2) The general reaction conditions were as follows: 35 gram lactulose was added to 35 gram 0.01 M sodium citrate buffer, pH 6.5. Subsequently, enzyme was dissolved in 10 ml water and added to initialize the reaction. Of most enzymes, 20 lactose units (LU)/gram lactulose were used. However, Lactase 14 DS and Tolerase 100 were used in a concentration of 200 lactose units (LU)/gram lactulose, due to their low activity at the general reaction conditions used. The reaction mixtures were placed in a water bath with an orbital shaker, thermostated at 50° C. After 48 hours reaction, the reaction was quenched by adding 1.5% 1M HCl and subsequently heated at 95° C. for 30 minutes.

(3) The fGOS content of the reaction mixtures was analyzed by HPLC (ThermoFisher Scientific Dionex type ICS 3000), based on the peak area percentage of individual sugars. The fGOS content was calculated by the following formula:

fGOS content (%_{ds})=100%–galactose %–glucose %–lactose %–lactulose %–fructose %.

(4) The fGOS content obtained with the different enzymes is summarized in Table 1. Table 1 shows that the enzyme obtained from *Papiliotrema terrestris* gave a far higher yield than the beta-galactosidase from *Aspergillus oryzae*; the best enzyme according to C. Guerrero et al.

(5) The ratio between the two building blocks of lactulose, namely fructose and galactose, is used as an indicator of the enzyme performance. When the ratio is <1, it indicates that the fGOS content achieved cannot be increased anymore by prolonging the reaction time, because it can only hydrolyze the substrate or fGOS formed, as is the case with *Aspergillus oryzae*-derived beta-galactosidase. For the *P. terrestris*-derived beta-galactosidase, the ratio of fructose/galactose is above 2, suggesting that the fGOS yield can be further optimized with prolonged reaction time.

(6) TABLE-US-00001 TABLE 1 fGOS composition formed by different beta-galactosidases

Enzyme	Enzyme origin	Gal	Glu	Fruc	lactulose	fGOS*	Fruc/Gal
Lactase 14-DS	<i>Aspergillus Oryzae</i>	23.68	2.51	24.54	11.86	37.41	1.04
Tolerase 100	<i>Aspergillus Oryzae</i>	28.61	2.85	26.74	7.55	34.25	0.93
Biolactase F	<i>Aspergillus Oryzae</i>	26.34	2.78	25.88	9.2	35.8	0.98
Amano PT	<i>Papiliotrema terrestris</i>	5.22	1.99	11.63	27.59	53.57	2.23

*Estimated by the peak percentage of the mono sugars and the substrate left.

(7) The HPLC chromatogram showed much more peaks for the fGOS produced by the *Papiliotrema terrestris*-derived enzyme compared to the fGOS produced with the *Aspergillus oryzae*-derived enzymes. Structure diversity of oligosaccharides is very important to serve the

nutritional needs of the different bifidobacteria in the gut and the decoy function for binding pathogens. Therefore, the fGOS derived from the process of the present invention seems of higher nutritional value.

(8) The degree of polymerization (DP) of the fGOS formed by the different enzymes is summarized in Table 2. Since it is not possible to distinguish lactulose from other DP2 fGOS components formed by lactulose, the total fGOS+lactulose content for each enzyme is given. Since lactulose itself is non-digestible (prebiotic) sugar, there is no need to separate it from fGOS. In the experiment with the *P. terrestris* enzyme, the fGOS+lactulose content is around 85%. In contrast, the fGOS+lactulose obtained from the *Aspergillus oryzae* enzymes is below 35%, which is probably due to the their intrinsic high hydrolytic activity as shown by their high mono sugar contents.

(9) TABLE-US-00002

TABLE 2 DP composition of fGOS formed by different beta-galactosidases	
Mono sugars	fGOS + Gal + Fru Glc DP2 DP3 DP4 DP5 DP6 Gal + Glu + Fru lactulose Lactase
14-DS	68.1 0.12 20.72 6.1 1.68 0.54 0.19 68.22 31.78 Biolactase F 75.38 0 18.73 3.68 0.71 0.14
0.03 75.38 24.62 Tolerase too	81.59 0.2 13.64 1.99 0.25 0 0 81.79 18.21 Amano PT 13.77 0.89
59.54 19.74 2.76 0.59 0 14.66 85.34	

(10) The experiment with *P. terrestris* was repeated with lactose as the substrate. The results are displayed in Table 3. It shows that the DP2-content of the fGOS (obtained from lactulose) is significantly higher than that of GOS obtained from lactose.

(11) This high DP2 fGOS component may constitute an advantage for the infant gut microbiota, because the growth of infant type of bifidobacteria such as *B. Breve* in the infant colon may be selectively promoted.

(12) TABLE-US-00003

TABLE 3 Comparison of Lactose-GOS and Lactulose GOS formed by <i>P. Terrestris</i> beta- galactosidase	
Mono sugars	Substrate Glc DP2 DP3 DP4 DP5 DP6 Gal + Glu + Fru
GOS + lactulose Lactulose	0.89 59.54 19.74 2.76 0.59 0 14.66 85.34 Lactose 20.04 39.3 24.45
11.35 2.26 0.23 22.23 60	

Claims

1. A process for the preparation of a galacto-oligosaccharide preparation, which process comprises the step of contacting a lactulose-containing feed with a beta-galactosidase (EC 3.2.1.23) derived from *Papiliotrema terrestris*.
2. The process according to claim 1 wherein the beta-galactosidase is derived from *Papiliotrema terrestris* strain MM13-F2171 (Accession Number: NITE BP-02177) or APC-6431 (Accession Number: NITE BP-02178).
3. The process according to claim 1 wherein the beta-galactosidase comprises an amino acid sequence according to any of SEQ ID NO: 1, 2, 3 or 4, or an amino acid sequence that is at least 80% identical to any of SEQ ID NO: 1, 2, 3 or 4.
4. The process according to claim 1 wherein the lactulose-containing feed comprises 40-58 wt. % lactulose.
5. The process according to claim 1 wherein the pH of the lactulose-containing feed is in the range 3.5-6.5.
6. The process according to claim 1 wherein the beta-galactosidase is used in a concentration of 1-20 LU/gram lactulose.
7. The process according to claim 1 wherein the lactulose is reacted with the beta-galactosidase at a temperature in the range 50-75° C.
8. A galacto-oligosaccharide preparation obtained by the process of claim 1.
9. A galacto-oligosaccharide preparation that is clinically free from lactose and which comprises, based on dry matter, 30-60 wt % of galacto-oligosaccharides other than lactulose.
10. The galacto-oligosaccharide preparation according to claim 9 comprising, based on dry matter,

40-70 wt % of a mixture of lactulose and monosugars.

11. A nutritional composition comprising the preparation according to claim 8, for administration to a subject known to suffer or having an increase chance to suffer from hypersensitivity to a GOS preparation obtained by transgalactosylation of lactose using a beta-galactosidase derived from *Bacillus circulans* or from *Aspergillus oryzae*.

12. A nutritional composition comprising the preparation according to claim 8, for administration to a subject known to suffer or having an increase chance to suffer from lactose intolerance.

13. A nutritional composition comprising the galacto-oligosaccharide preparation according to claim 8 and at least one further ingredient selected from the group consisting of a protein source, probiotics, LC-PUFA's and carbohydrates.

14. The nutritional composition according to claim 13 wherein said composition is suitable for consumption by adults and children aged at least 18 months.
